Semantic Margin-Tightened MPC-SECBF for Context-Dependent Dynamic Obstacle Avoidance

Zhang Bo, Lu Xuran

***Abstract*—This paper addresses context-dependent dynamic obstacle avoidance for autonomous mobile robots in social dynamic environments, where safety requirements vary with obstacle category and interaction state, including head-on approach, short-term collision urgency, and local crowding. A semantic margin-tightened MPC-ECBF framework is therefore proposed to incorporate category- and interaction-aware safety requirements into velocity-aware dynamic obstacle avoidance through a bounded semantic safety margin. Within this framework, a semantic extended Euclidean safety metric (SEESM) is constructed by preserving the dynamic geometric risk kernel of the extended Euclidean safety metric and tightening it with the bounded semantic safety margin. The semantic margin is then computed from these category and interaction-risk factors. To prevent infeasible or overly conservative boundary tightening, the semantic margin is constrained by category-dependent upper bounds, positive-increment limits, the currently available dynamic geometric safety margin, and an MPC feasibility guard. The resulting SEESM is embedded into a hierarchical planning framework, where it is used for global motion-primitive collision checking and for constructing semantic extended control barrier function constraints in local MPC. A closed-loop practical safety lower bound is derived to quantify the effects of bounded CBF relaxation and semantic-margin updates. Component-level tests, simulation benchmarking, and real-world experiments validate the proposed MPC-SECBF framework in terms of semantic safety clearance, feasibility preservation, and real-time dynamic obstacle avoidance.**

***Index Terms*—Autonomous mobile robots, context-dependent dynamic obstacle avoidance, semantic margin tightening, model predictive control, control barrier function.**

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I. Introduction

In social dynamic environments, avoidance behavior is inherently context-dependent. Humans do not determine safe passing distances solely from geometric proximity; they also consider the type of obstacle and the evolving interaction condition. For example, a fast approaching cyclist, a child in a crowded area, and a static box imply different safety requirements even under similar geometric distances. This motivates context-dependent dynamic obstacle avoidance for autonomous mobile robots, in which safety boundaries should adapt to obstacle category and interaction state.

Existing dynamic obstacle avoidance methods have provided effective tools for modeling motion-induced collision risk. Prediction-aware and risk-aware planners reason about future obstacle occupancy, relative motion, and uncertainty to support proactive collision avoidance in dynamic scenes In parallel, MPC-CBF and dynamic CBF formulations embed safety constraints into predictive trajectory optimization, providing a principled mechanism to balance tracking performance, feasibility, and collision avoidance Recent adaptive-margin CBF methods further show that safety margins can be adjusted according to interaction uncertainty or human-related factors. These advances establish a strong foundation for safety-critical navigation in dynamic environments. In social dynamic environments, this foundation can be further extended by allowing the safety boundary to reflect category- and interaction-dependent requirements, rather than being determined only by geometric size and kinematic state.

Semantic and social safety-aware navigation has recently become an important direction for extending geometric obstacle avoidance toward human-centered environments. One line of work incorporates semantic risk into spatial safety representations. Semantic labels have been used to inflate occupied regions, reshape distance fields, and tune barrier-function gains, so that semantically sensitive objects induce more conservative safety responses than ordinary obstaclesAnother line of work constructs semantic safety functions or guidance fields from perception-level semantics. Social-semantic safety filters, Poisson safety functions, Laplace guidance fields, and language-conditioned safety filters translate semantic labels, contextual rules, or natural-language safety specifications into differentiable safety functions or controller constraintsBeyond semantic safety representation, social navigation studies have explored how interaction preferences can be incorporated into planning and control. Learned social zones and proxemic constraints can be encoded as barrier functions or cost terms to enforce human-like clearance, yielding, and personal-space preservationTopological and homotopy-based planning methods represent passing-side preferences through winding numbers, homotopy classes, or social costs, enabling robots to select socially compliant paths under dynamic pedestrian interactionsRecent adaptive-margin, trust-aware, and uncertainty-aware CBF/MPC methods further suggest that safety margins need not be fixed, but can be adjusted according to interaction uncertainty, human-related factors, prediction confidence, or online risk estimatesThese studies demonstrate that semantic categories, social norms, and interaction context can be connected to low-level safety control. Nevertheless, existing formulations usually follow one of three routes: reshaping a spatial safety field, enforcing social zones as hard constraints, or adding social preferences at the planning layer. For velocity-aware dynamic obstacle avoidance, it remains insufficiently explored how category- and interaction-dependent safety requirements can be incorporated into an MPC-ECBF formulation as a bounded tightening of the dynamic safety margin, while preserving the physical interpretation of the EESM dynamic-risk kernel and keeping soft passing preferences separate from hard safety constraints.

Motivated by the above observations, this paper proposes a semantic margin-tightened MPC-ECBF framework for context-dependent dynamic obstacle avoidance. The key idea is to preserve the velocity-aware dynamic risk kernel of EESM and to introduce semantic information only through a bounded margin tightening term. Specifically, a semantic safety margin is generated from obstacle category and interaction-risk factors, including head-on approach, short-term collision urgency, and local crowding. The margin is further constrained by category-dependent upper bounds, positive-increment limits, the currently available dynamic geometric safety margin, and an MPC feasibility guard. In this way, semantic risk is incorporated into the near-horizon ECBF constraints without modifying the physical interpretation of the EESM kernel. Side-passing preference is introduced only as a soft cost in the MPC objective, so that it guides socially consistent lateral behavior without changing the hard feasible set.

The main contributions of this paper are summarized as follows:

1) A semantic margin-tightened MPC-ECBF framework is developed for context-dependent dynamic obstacle avoidance, where category- and interaction-aware safety requirements are incorporated into velocity-aware obstacle avoidance while preserving the EESM dynamic-risk kernel.
2) A guarded semantic-margin generation and update mechanism is proposed to prevent infeasible or overly conservative boundary tightening by enforcing category-dependent upper bounds, positive-increment limits, available-margin projection, and MPC feasibility checking.
3) Theoretical properties of the proposed formulation are derived to make the semantic tightening analyzable and compatible with MPC feasibility, including degeneration consistency, margin-shift equivalence, a feasibility sufficient condition, and a closed-loop practical safety lower bound under bounded CBF relaxation and semantic-margin updates. These results characterize the practical safety and feasibility effects of semantic tightening without claiming full recursive feasibility or unconditional safety under arbitrary semantic perception errors.
4) Component-level tests, simulation benchmarking, and real-world experiments are conducted to evaluate semantic safety clearance, feasibility preservation, and real-time dynamic obstacle avoidance performance.

The remainder of this paper is organized as follows. Section II presents the proposed semantic margin-tightened MPC-ECBF framework, including the construction of the semantic safety margin, the SEESM-based safety function, the guarded update mechanism, and the MPC-SECBF formulation. Section III provides the theoretical analysis of the proposed formulation, including feasibility and practical safety properties. Section IV reports component-level validation, simulation benchmarking, and real-world experiments. Section V concludes this paper and discusses future work.

II. Problem Formulation

This work considers context-dependent dynamic obstacle avoidance for autonomous mobile robots operating in social dynamic environments. At each sampling time t, the context-dependent dynamic obstacle avoidance task is cast as a finite-horizon MPC problem $\mathcal{P}_t$:

$$
\begin{aligned}
\min_{\mathbf{x}_t,\mathbf{u}_t,\boldsymbol{\epsilon}_t} \quad & J_t + \sum_{k=0}^{N_s-1}\sum_{i=1}^{n_t} \rho_\epsilon \epsilon_{i,t+k}^2 \\
\text{s.t.} \quad & x_{t|t} = x_t, \\
& x_{t+k+1|t} = f_d(x_{t+k|t}, u_{t+k|t}), \\
& u_{t+k|t} \in \mathcal{U}, \quad x_{t+k|t} \in \mathcal{X}, \\
& 0 \le \epsilon_{i,t+k} \le \epsilon_{\max}, \\
& h_i^{\mathrm{ctx}}(X_{t+k+1|t};\eta_{i,t}) \ge (1-\gamma) h_i^{\mathrm{ctx}}(X_{t+k|t};\eta_{i,t}) \\
& \qquad - \epsilon_{i,t+k},
\end{aligned} \tag{1}
$$

where $k = 0,\dots,N_s-1$ and $i = 1,\dots,n_t$. Here, J_t penalizes tracking error and control effort, $\mathcal{U}$ and $\mathcal{X}$ denote the admissible input and state sets, and $h_i^{\mathrm{ctx}}(\cdot;\eta_{i,t})$ denotes a context-dependent safety function parameterized by the safety-requirement variable $\eta_{i,t}$. The variable $\eta_{i,t}$ is used here only to represent the fact that the required safety boundary may vary with obstacle category and interaction state; its concrete construction is developed in the following sections. The relaxation variable $\epsilon_{i,t+k}$ accounts for limited control authority and abrupt environmental changes, so the resulting safety property is interpreted as practical safety rather than unconditional hard safety.

The input information includes the robot state, dynamic obstacle positions and velocities, short-horizon obstacle predictions, semantic categories, and interaction states. These quantities are assumed to be available from perception and tracking modules or from experimental annotations. Semantic recognition, long-term human intent prediction, and social behavior learning are outside the scope of this paper. Under these assumptions, the research problem is to construct $h_i^{\mathrm{ctx}}(\cdot;\eta_{i,t})$ and an admissible update rule for $\eta_{i,t}$, such that category- and interaction-dependent safety requirements can be incorporated into velocity-aware MPC-ECBF while avoiding infeasible or overly conservative boundary tightening and maintaining real-time implementation.

III. Main Method

A. Framework Overview

Fig. **??** illustrates the overall architecture of the proposed semantic margin-tightened MPC-SECBF framework. At each sampling time, RGB-D images, point clouds, and IMU data are used to provide obstacle categories, obstacle states, short-horizon trajectory predictions, and robot odometry. The obstacle category $c_{i,t}$ is used to determine a category prior margin, while the predicted obstacle motion is used to evaluate interaction-risk features, including head-on approach, short-term collision urgency, and local crowding. These two sources are combined to generate a context-aware candidate semantic margin.

The candidate margin is then passed through a feasibility-preserving margin update module. This module enforces category-dependent upper bounds, positive-increment limits, available-margin constraints, and an MPC feasibility guard, yielding a bounded and feasible margin. The resulting margin is used to construct the SEESM safety function by tightening the EESM velocity-aware safety boundary. In parallel, SLAM and global planning provide the robot state estimate and a global reference path. The SEESM safety function is embedded as SECBF constraints in the local MPC optimizer, which tracks the reference path and generates the control command for the autonomous mobile robot.

The proposed framework instantiates the abstract problem in Section **??** through three key steps. First, the context-dependent safety function $h_i^{\mathrm{ctx}}(\cdot;\eta_{i,t})$ is realized as the SEESM safety function, which preserves the velocity-aware EESM dynamic-risk kernel while introducing semantic boundary tightening. Second, the abstract safety-requirement parameter $\eta_{i,t}$ is specified as the semantic margin $\beta_{i,t}$, whose value is determined by obstacle category and interaction-risk factors. Third, the admissible update of $\eta_{i,t}$ is implemented as a feasibility-preserving semantic-margin update, which bounds margin variation and prevents infeasible or overly conservative tightening before the resulting SEESM constraint is embedded into MPC-ECBF.

B. Semantic Extended Euclidean Safety Metric

C. Feasibility-Preserving Semantic Margin Update

D. MPC-SECBF Formulation with Soft Passing Preference

E. Framework Overview

F. Dynamic Perception

IV. Semantic Extended Euclidean Safety Margin

A. Dynamic Geometric Safety Distance

In motion planning, the robot and obstacles are commonly abstracted as circular regions. Let R_r and R_o denote the robot and obstacle radii, respectively. The inflated radius of obstacle i is denoted by $R_i = R_r + R_o$, where $i \in \{1,\ldots,n\}$. The conventional Euclidean safety margin (CESM) evaluates safety only from the instantaneous geometric distance between the robot and obstacles, and defines the safe region as

$$SR_{\mathrm{CE}} = \{p(t) \mid \|p(t) - p_i^o(t)\|_2 > R_d,\ \forall i\}, \tag{2}$$

where $R_d = R_i + R_{\mathrm{safe}}$, and R_{safe} is the prescribed safety threshold. Here, $p(t) = [p_x(t), p_y(t)]^\top$ denotes the robot position, $p_i^o(t)$ denotes the position of obstacle i at time t, and $\|p(t) - p_i^o(t)\|_2$ denotes their Euclidean distance. According to CESM, the current state is regarded as unsafe when the relative distance to any obstacle is smaller than R_d.

To describe short-horizon interaction risk induced by dynamic obstacles, the velocity-extended idea in EESM is introduced. The dynamic geometric term is composed of relative position, relative velocity, and forward integration time. The relative position and relative velocity between the robot and obstacle i are

$$l_i = p(t) - p_i^o(t), \qquad v_i = v(t) - v_i^o(t). \tag{3}$$

The corresponding basic dynamic geometric safe region is

$$SR_{\mathrm{EE}} = \{p(t) \mid \|l_i + \tau_i v_i\|_2 > R_d,\ \forall i\}\,. \tag{4}$$

where $v(t) = [v_x(t), v_y(t)]^\top$ is the robot velocity, $v_i^o(t)$ is the velocity of obstacle i, and τ_i is the forward integration time along the relative velocity direction. The vector $l_i + \tau_i v_i$ represents the extended geometric displacement under short-term relative motion, and $\|l_i + \tau_i v_i\|_2$ quantifies the dynamic geometric distance. The angle δ_i denotes the angle between l_i and v_i. Specifically, the forward integration time is defined as

$$\tau_i = f_r f_v f_T K_e T_i, \qquad K_e \in (0,1), \tag{5}$$

where

$$T_i = \frac{(\|l_i\|_2 - R_i)\cos\delta_i}{\|v_i\|_2}, \tag{6}$$

$$f_r = \mathrm{sign}(-\cos\delta_i), \tag{7}$$

$$f_v = \mathrm{sign}\left((l_i^\top v_i)^2 + (R_i^2 - \|l_i\|_2^2)\|v_i\|_2^2\right), \tag{8}$$

and

$$f_T = \mathrm{sign}(T_{\max} - T_i). \tag{9}$$

Here, T_i estimates the time for the robot to reach the obstacle collision boundary; f_r is a relative-direction factor that checks whether the robot and obstacle are moving toward each other; f_v is a velocity-obstacle factor that checks whether the relative

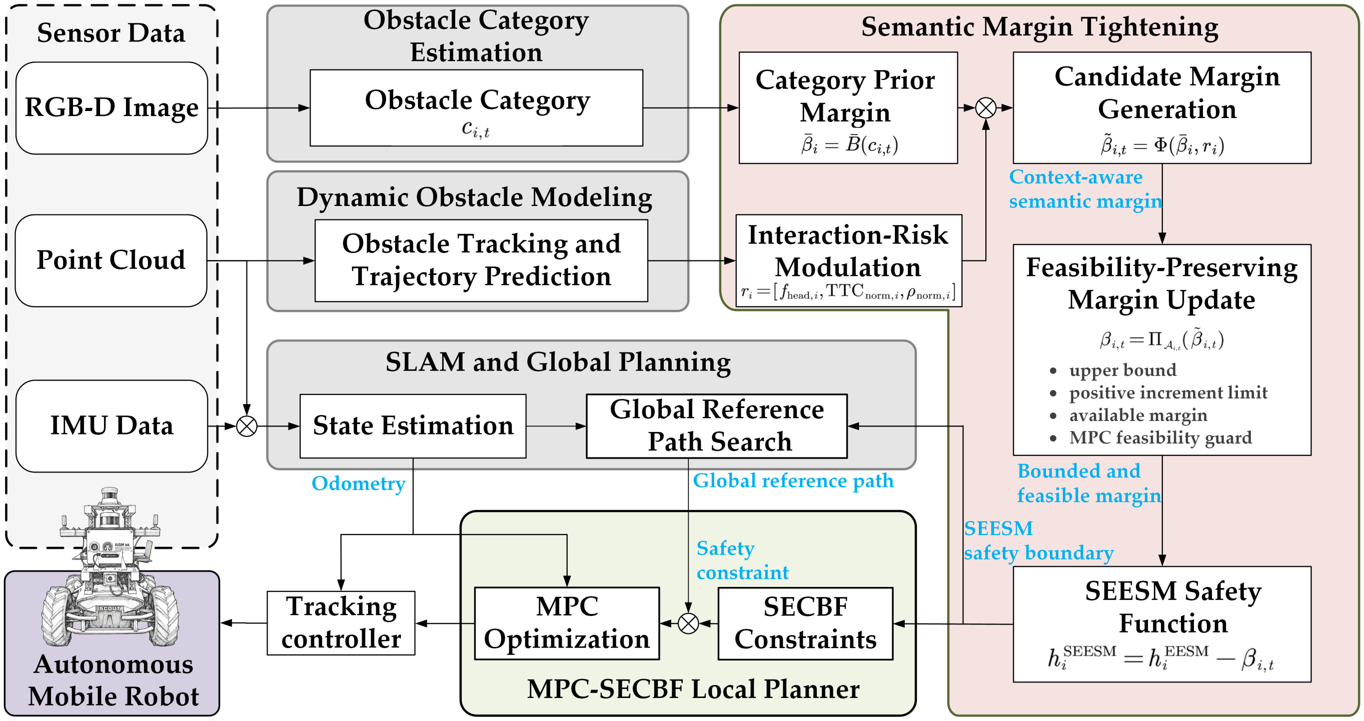

Fig. 1. **Semantic Margin-Tightened MPC-SECBF Framework**.

velocity lies in the collision cone; f_T is a time-horizon factor that checks whether the potential collision lies within the prediction horizon; and K_e relaxes the influence of the forward integration time on the safety distance. The indicator function is

$$\text{sign}(x) = \begin{cases} 1, & x > 0, \\ 0, & x \le 0. \end{cases} \tag{10}$$

Therefore, the dynamic geometric distance and the corresponding safety function are

$$d_{\text{EE}}(X_t) = \|l_i + \tau_i v_i\|_2, \tag{11}$$

and

$$h_{\text{EE}}(X_t) = d_{\text{EE}}(X_t) - R_d = \|l_i + \tau_i v_i\|_2 - R_d. \tag{12}$$

When $h_{\text{EE}}(X_t) > 0$, the robot remains safe with respect to obstacle i in the dynamic geometric sense. When $h_{\text{EE}}(X_t) \le 0$, the current state is judged to have potential collision risk.

B. Semantic Safety Margin Generation

In shared social navigation environments, different obstacle categories have different safety sensitivities. For example, children and cyclists usually carry higher uncertainty and interaction risk, while non-social objects such as boxes require a smaller additional margin. Therefore, a semantic safety margin $\beta_{i,t}$ is introduced for each obstacle to tighten the basic dynamic geometric boundary.

The maximum available semantic margin is determined by the semantic category:

$$\bar{\beta}_i = \bar{B}(c_{i,t}), \tag{13}$$

where $c_{i,t}$ is the semantic category of obstacle i at time t. The category mapping is

$$\bar{B}(c_{i,t}) = \begin{cases} 0.7\,\text{m}, & c_{i,t} = \text{child}, \\ 0.6\,\text{m}, & c_{i,t} = \text{cyclist}, \\ 0.5\,\text{m}, & c_{i,t} = \text{vehicle}, \\ 0.4\,\text{m}, & c_{i,t} = \text{pedestrian}, \\ 0.4\,\text{m}, & c_{i,t} = \text{unknown}, \\ 0.1\,\text{m}, & c_{i,t} = \text{box}. \end{cases} \tag{14}$$

Because category alone cannot capture the risk variation within the same class, a context-risk modulation mechanism is further introduced. The context risk intensity of obstacle i is

$$r_i = \text{clip}\left(\lambda_h f_{\text{head},i} + \lambda_T TTC_{\text{norm},i} + \lambda_\rho \rho_{\text{norm},i}, 0, 1\right). \tag{15}$$

where $f_{\text{head},i}$, $TTC_{\text{norm},i}$, and $\rho_{\text{norm},i}$ denote the head-on approach degree, short-term collision urgency, and local crowding degree, respectively; and λ_h, λ_T, and λ_ρ are the corresponding weights. The head-on feature is

$$f_{\text{head},i} = \max(0, -\cos\delta_i). \tag{16}$$

This term is a continuous counterpart of the relative-direction indicator $f_r = \text{sign}(-\cos\delta_i)$. The two terms play different roles: f_r determines whether to enable the forward integration time τ_i in the dynamic geometric distance, whereas $f_{\text{head},i}$ continuously describes the head-on approach degree for semantic-margin modulation.

To quantify short-term collision urgency, the closing speed is defined as

$$v_i^{\mathrm{clo}} = \left[-\frac{l_i^\top v_i}{\|l_i\|_2 + \epsilon_l} \right]_+ , \tag{17}$$

where $[a]_+ = \max(a, 0)$ and $\epsilon_l > 0$ avoids division by zero. The approximate time to collision is then

$$TTC_i = \frac{[\|l_i\|_2 - R_i]_+}{v_i^{\mathrm{clo}} + \epsilon_v}, \tag{18}$$

and its normalized risk intensity is

$$TTC_{\mathrm{norm},i} = \mathrm{clip}\left(1 - \frac{TTC_i}{TTC_{\max}}, 0, 1\right). \tag{19}$$

A smaller TTC_i makes $TTC_{\mathrm{norm},i}$ closer to one, indicating higher short-term collision risk, whereas a larger TTC_i drives this term toward zero. The TTC_i term is used only for context-risk modulation and does not replace T_i or τ_i in the dynamic geometric safety distance.

Local density describes the crowding degree around an obstacle. Given a neighborhood radius R_ρ, the number of neighboring obstacles around obstacle i is

$$N_i^\rho = \sum_{j \neq i} \mathbb{I}\left(\|p_j^o(t) - p_i^o(t)\|_2 \leq R_\rho\right), \tag{20}$$

$$\rho_i = \frac{N_i^\rho}{\pi R_\rho^2}, \qquad \rho_{\mathrm{norm},i} = \mathrm{clip}\left(\frac{\rho_i}{\rho_{\max}}, 0, 1\right). \tag{21}$$

The corresponding local density and normalized local density are denoted by ρ_i and $\rho_{\mathrm{norm},i}$, respectively. The weights satisfy

$$\lambda_h \geq 0, \qquad \lambda_T \geq 0, \qquad \lambda_\rho \geq 0, \qquad \lambda_h + \lambda_T + \lambda_\rho = 1. \tag{22}$$

Based on the category upper bound and the context risk, the raw candidate semantic margin is

$$\tilde{\beta}_{i,t} = \bar{\beta}_i \left[\mu_{\min} + (1 - \mu_{\min}) r_i\right], \tag{23}$$

where $\mu_{\min} \in [0, 1]$ is the minimum activation ratio. To avoid abrupt tightening of the semantic boundary between adjacent sampling instants, the positive rate of change is limited by

$$\hat{\beta}_{i,t} = \min\left\{\tilde{\beta}_{i,t}, \beta_{i,t-1} + \bar{\Delta}_{\beta,i}\right\}, \tag{24}$$

which implies

$$[\hat{\beta}_{i,t} - \beta_{i,t-1}]_+ \leq \bar{\Delta}_{\beta,i}. \tag{25}$$

To ensure that the semantic margin does not exceed the available physical safety margin, the admissible semantic set is defined as

$$\mathcal{B}_i(X_t) = \left\{\beta \in [0, \bar{\beta}_i] \mid h_{\mathrm{EE}}(X_t) - \beta \geq 0\right\}. \tag{26}$$

Equivalently,

$$\mathcal{B}_i(X_t) = [0, \bar{\beta}_i] \cap (-\infty, h_{\mathrm{EE}}(X_t)]. \tag{27}$$

When $h_{\mathrm{EE}}(X_t) \geq 0$, this set becomes

$$\mathcal{B}_i(X_t) = \left[0, \min\{\bar{\beta}_i, h_{\mathrm{EE}}(X_t)\}\right]. \tag{28}$$

The rate-limited candidate is projected onto the admissible set:

$$\beta_{i,t}^{\mathrm{safe}} = \Pi_{\mathcal{B}_i(X_t)}(\hat{\beta}_{i,t}). \tag{29}$$

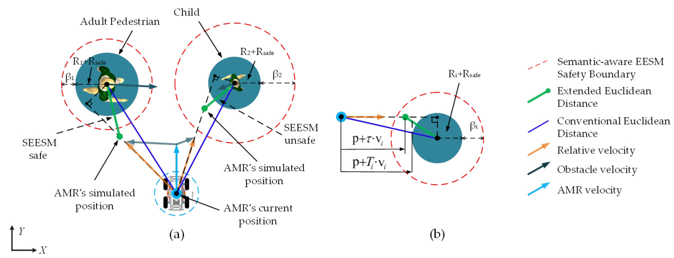

Fig. 2. Principle of the proposed SEESM. The semantic margin $\beta_{i,t}$ tightens the dynamic Euclidean safety boundary according to obstacle category and context risk, producing a semantic safety boundary used by both global collision checking and local MPC-SECBF constraints.

The semantic margin used by the MPC-SECBF constraints is

$$\beta_{i,t} = \begin{cases} \beta_{i,t}^{\mathrm{safe}}, & \mathcal{P}_t^{\mathrm{sem}}(\beta_{i,t}^{\mathrm{safe}}) \text{ is feasible}, \\ \Pi_{\mathcal{B}_i(X_t)}(\beta_{i,t-1}), & \text{otherwise}. \end{cases} \tag{30}$$

Thus, the final semantic safety margin is not simply the candidate $\hat{\beta}_{i,t}$, but is obtained under the category upper bound, context risk, positive-increment limit, current safety margin, and MPC feasibility guard.

C. Semantic Extended Euclidean Safety Metric

Given $\beta_{i,t}$, the semantic extended Euclidean safety function is

$$h_{\mathrm{SEE}}(X_t; \beta_{i,t}) = h_{\mathrm{EE}}(X_t) - \beta_{i,t}, \tag{31}$$

Substituting the definition of h_{EE} gives

$$h_{\mathrm{SEE}}(X_t; \beta_{i,t}) = \|l_i + \tau_i v_i\|_2 - R_d - \beta_{i,t}. \tag{32}$$

The semantic extended Euclidean safe region is

$$SR_{\mathrm{SEE}} = \{p(t) \mid \|l_i + \tau_i v_i\|_2 > R_d + \beta_{i,t}, \ \forall i\}. \tag{33}$$

Thus, SEESM does not directly change the physical radius of the robot or obstacle. Instead, it adds a semantic margin $\beta_{i,t}$ to the basic dynamic geometric boundary R_d. The semantic collision indicator can be written as

$$F_{\mathrm{SEE}} = \|l_i + \tau_i v_i\|_2 - (R_d + \beta_{i,t}). \tag{34}$$

D. SEESM-Based Path Search

During primitive expansion, the method samples the control input and duration, and then forward-integrates the system to obtain candidate motion primitives. The execution cost is jointly determined by the control input $u(t)$ and the primitive duration T:

$$f_e(t) = \int_0^T \|u(t)\|_2^2 \, dt + \rho T, \tag{35}$$

where ρ is a time-penalty weight that suppresses excessively long primitives. The heuristic cost f_h is obtained by solving a two-point boundary-value problem to estimate the shortest-distance cost from the current state to the goal.

During collision checking, the semantic safety margin is introduced on top of the dynamic geometric safety test, resulting in CheckSEESM$(\cdot)$. For candidate primitive node

m and predicted obstacle i, let l_{mi}, v_{mi}, and τ_{mi} denote the relative position, relative velocity, and forward integration time, respectively. If

$$C_{mi} : \|l_{mi} + \tau_{mi} v_{mi}\|_2 - (R_{d,i} + \beta_{i,t}) \le 0. \tag{36}$$

then the primitive node is regarded as unsafe. This criterion indicates that path search accounts not only for the relative motion trend of dynamic obstacles, but also for the semantic safety margin $\beta_{i,t}$ determined by category and context risk. Therefore, unsafe primitives that violate the semantic safety constraint can be removed before trajectory optimization.

E. Semantic Extended Control Barrier Function

To incorporate CBF constraints into dynamic obstacle avoidance, SECBF uses SEESM to construct the barrier function and introduces slack variables to balance safety and feasibility. Let $\mathcal{Z} = \{X \in D \mid h(X) \ge 0\}$ be the safe set, where $h : D \to \mathbb{R}$ is continuously differentiable. For the input-affine system $\dot{x} = f(x) + g(x)u$, if there exists an extended class-$\mathcal{K}$ function $\gamma(\cdot)$, the continuous-time CBF condition can be written as

$$\sup_{u \in U} \left[\dot{h}(X, u)\right] \ge -\gamma(h(X)), \qquad \gamma \in K_\infty, \tag{37}$$

with

$$\dot{h}(X, u) = L_f h(X) + L_g h(X) u + \frac{\partial h}{\partial p^o} \frac{\partial p^o}{\partial t}. \tag{38}$$

where U denotes the feasible input set subject to physical and operational limits, and $L_f h(X)$ and $L_g h(X)$ are the Lie derivatives of $h(X)$ along the drift term $f(x)$ and control input term $g(x)$, respectively. Compared with standard DCBF-based CBF design, the additional term $\partial h / \partial p^o$ captures the effect of dynamic obstacle-state variation on the safety-function derivative.

To include CBF constraints in a discrete-time controller, the derivative is approximated by the one-step difference $\dot{h}(X_k) \approx [h(X_{k+1}) - h(X_k)]/\Delta t$. The relaxed discrete-time CBF recursion is then

$$\begin{aligned} h(X_{k+1}) &\ge (1-\gamma) h(X_k) - \epsilon_k, \\ &0 < \gamma < 1, \quad 0 \le \epsilon_k \le \epsilon_{\max}. \end{aligned} \tag{39}$$

where γ is treated as a scalar for compact discrete-time notation, and ϵ_k is a nonnegative slack variable that relaxes the CBF constraint within a bounded range to improve optimization feasibility. Starting from $k = 0$, recursion gives

$$h(X_k) \ge (1-\gamma)^k h(X_0) - \sum_{i=0}^{k-1} (1-\gamma)^{k-1-i} \epsilon_i. \tag{40}$$

Since $\epsilon_i \le \epsilon_{\max}$,

$$\sum_{i=0}^{k-1} (1-\gamma)^{k-1-i} \epsilon_i \le \epsilon_{\max} \sum_{j=0}^{k-1} (1-\gamma)^j, \tag{41}$$

and

$$\sum_{j=0}^{k-1} (1-\gamma)^j = \frac{1 - (1-\gamma)^k}{\gamma}. \tag{42}$$

Therefore,

$$h(X_k) \ge (1-\gamma)^k h(X_0) - \epsilon_{\max} \frac{1 - (1-\gamma)^k}{\gamma}. \tag{43}$$

If $X_0 \in \mathcal{Z}$, namely $h(X_0) \ge 0$, then

$$h(X_k) \ge -\epsilon_{\max} \frac{1 - (1-\gamma)^k}{\gamma} \triangleq -\kappa_k, \tag{44}$$

where κ_k is the instantaneous safety-deviation bound at step k. As k increases, κ_k monotonically approaches $\kappa_\infty = \epsilon_{\max}/\gamma$. Consequently,

$$h(X_k) \ge -\frac{\epsilon_{\max}}{\gamma}. \tag{45}$$

To keep the practical safety deviation within the prescribed safety threshold, the parameters can be selected to satisfy $\epsilon_{\max} \le \gamma R_{\text{safe}}$.

In dynamic obstacle avoidance, the construction of $h(\cdot)$ directly determines the effectiveness of the CBF constraint. Considering the predicted obstacle trajectory over the MPC horizon, let k denote a discrete prediction step. The parameterized state of obstacle i at step k is $[p_{x,i}(k), p_{y,i}(k), v_{x,i}(k), v_{y,i}(k), R_i]^\top$. Based on CESM, the dynamic geometric safety distance, and SEESM, the safety functions are

$$h_{\text{CE}}(X_k) = \|l_i(k)\|_2 - R_i - R_{\text{safe}}, \quad k = 0, 1, \dots, N-1, \tag{46}$$

$$\begin{aligned} h_{\text{EE}}(X_k) &= \|l_i(k) + \tau_i v_i(k)\|_2 - R_i - R_{\text{safe}}, \\ &k = 0, 1, \dots, N-1, \end{aligned} \tag{47}$$

and

$$\begin{aligned} h_{\text{SEE}}(X_k; \beta_{i,t}) &= \|l_i(k) + \tau_i v_i(k)\|_2 - R_i - R_{\text{safe}} - \beta_{i,t}, \\ &k = 0, 1, \dots, N-1. \end{aligned} \tag{48}$$

F. Model Predictive Control

The trajectory tracking problem is formulated as an MPC optimization task that minimizes tracking error while preserving safety and feasibility. The robot state and input are

$$x_t = \begin{bmatrix} p_x(t) \\ p_y(t) \\ \theta(t) \end{bmatrix}, \qquad u_t = \begin{bmatrix} v(t) \\ \omega(t) \end{bmatrix}, \tag{49}$$

where $\theta(t)$ is the heading angle, and $v(t)$ and $\omega(t)$ are the linear and angular velocities. The discrete-time kinematic model is

$$x_{t+1} = x_t + \begin{bmatrix} \cos\theta(t) & 0 \\ \sin\theta(t) & 0 \\ 0 & 1 \end{bmatrix} u_t \Delta t. \tag{50}$$

Let $U_t = [u_{t|t}, u_{t+1|t}, \dots, u_{t+N-1|t}]$ and $\Psi_t = \{\epsilon_{i,k}\}$. The semantic MPC-SECBF problem is

$$\begin{aligned} \mathcal{P}_t^{\text{sem}} : \quad \min_{U,\Psi} \quad & p(x_{t+N|t}) + \sum_{k=0}^{N-1} q(x_{t+k|t}, u_{t+k|t}) \\ & + \sum_{i=1}^{n} \sum_{k=0}^{N-1} \psi(\epsilon_{i,k}) + J_t^{\text{side}}, \end{aligned} \tag{51}$$

where

$$J_t^{\text{side}} = \sum_{i=1}^{n} \sum_{k=0}^{N_s-1} w_{i,t}^{\text{side}} \phi(-g_{i,k}^{\text{side}}). \tag{52}$$

The optimization is subject to

$$x_{t+k+1|t} = f(x_{t+k|t}, u_{t+k|t}), \quad k = 0, \dots, N-1, \tag{53}$$

$$x_{t+k|t} \in X_{\text{tra}}^k, \quad u_{t+k|t} \in U, \quad k = 0, \dots, N-1, \tag{54}$$

$$x_{t|t} = x_t, \qquad x_{t+N|t} \in X_f, \tag{55}$$

$$\begin{aligned} h_{\text{SEE}}(X_{t+k+1|t}; \beta_{i,t}) &\geq (1-\gamma) h_{\text{SEE}}(X_{t+k|t}; \beta_{i,t}) - \epsilon_{i,k}, \\ k &= 0, 1, \dots, K_c N - 1, \end{aligned} \tag{56}$$

and

$$\begin{aligned} h_{\text{CE}}(X_{t+k+1|t}) &\geq (1-\gamma) h_{\text{CE}}(X_{t+k|t}) - \epsilon_k, \\ k &= K_c N, \dots, N-1. \end{aligned} \tag{57}$$

The cost terms are

$$p(x_{t+N|t}) = \|x_{t+N|t} - x_{t+N|t}^d\|_P, \tag{58}$$

$$\begin{aligned} q(x_{t+k|t}, u_{t+k|t}) = &\|x_{t+k|t} - x_{t+k|t}^d\|_Q + \|u_{t+k|t}\|_R \\ &+ \|u_{t+k+1|t} - u_{t+k|t}\|_S, \end{aligned} \tag{59}$$

and

$$\psi(\epsilon_{i,k}) = \lambda_\epsilon \epsilon_{i,k}^2. \tag{60}$$

The objective in $\mathcal{P}_t^{\text{sem}}$ consists of terminal tracking error, stage tracking and control costs, CBF slack penalties, and the side-passing preference cost. The matrices P, Q, R, and S are positive definite weights, while λ_ϵ penalizes slack variables. The term J_t^{side} guides more stable lateral passing behavior in the presence of semantically sensitive obstacles. Constraints on dynamics, state feasibility, control feasibility, initial state, terminal set, near-field SECBF, and long-horizon CESM-based DCBF are specified above. The stricter SECBF condition is used in the near field because short-term safety is critical for real-time collision avoidance, while the longer prediction horizon uses the more relaxed CESM-based DCBF condition to reduce conservatism under prediction uncertainty.

G. Closed-Loop Practical Safety

The preceding analysis assumes that the semantic safety margin is fixed within a single MPC prediction horizon. In closed-loop execution, however, $\beta_{i,t}$ is updated at different sampling instants according to perception and context risk, thereby changing the threshold of the safety function. This section quantifies the practical safety lower bound of the semantic-enhanced system.

1) Semantic Safety Set: Semantic enhancement cannot create safety by itself; it can only consume the dynamic geometric safety margin currently available. For any obstacle i, define the admissible semantic safety set at state X_t as

$$\mathcal{B}_i(X_t) = \{\beta \in [0, \bar{\beta}_i] : h_{\text{SEE}}(X_t; \beta) \geq 0\}. \tag{61}$$

Since

$$h_{\text{SEE}}(X_t; \beta) = h_{\text{EE}}(X_t) - \beta, \tag{62}$$

we have

$$h_{\text{SEE}}(X_t; \beta) \geq 0 \iff h_{\text{EE}}(X_t) - \beta \geq 0 \iff \beta \leq h_{\text{EE}}(X_t). \tag{63}$$

Together with $\beta \in [0, \bar{\beta}_i]$, this yields

$$\begin{aligned} \mathcal{B}_i(X_t) &= [0, \bar{\beta}_i] \cap (-\infty, h_{\text{EE}}(X_t)] \\ &= [0, \min\{\bar{\beta}_i, h_{\text{EE}}(X_t)\}]. \end{aligned} \tag{64}$$

When $h_{\text{EE}}(X_t) \geq 0$, this interval at least contains $\beta = 0$, and thus $\mathcal{B}_i(X_t) \neq \varnothing$. This shows that the optional semantic margin is directly bounded by the current dynamic geometric safety margin, providing a feasibility guard for semantic updates.

2) Closed-Loop Practical Safety: Define the closed-loop safety quantity

$$H_{i,t} := h_{\text{SEE}}(X_t; \beta_{i,t}). \tag{65}$$

Since $\beta_{i,t}$ is frozen during the MPC solve at time t, the first executed step satisfies

$$h_{\text{SEE}}(X_{t+1}; \beta_{i,t}) \geq (1-\gamma) h_{\text{SEE}}(X_t; \beta_{i,t}) - \epsilon_{i,t}. \tag{66}$$

Moreover,

$$\begin{aligned} H_{i,t+1} &= h_{\text{SEE}}(X_{t+1}; \beta_{i,t+1}) \\ &= h_{\text{SEE}}(X_{t+1}; \beta_{i,t}) - (\beta_{i,t+1} - \beta_{i,t}). \end{aligned} \tag{67}$$

With

$$\Delta\beta_i(t) := [\beta_{i,t+1} - \beta_{i,t}]_+, \tag{68}$$

it follows that

$$H_{i,t+1} \geq (1-\gamma) H_{i,t} - \epsilon_{i,t} - \Delta\beta_i(t). \tag{69}$$

Expanding the recursion gives

$$H_{i,t} \geq (1-\gamma)^t H_{i,0} - \sum_{j=0}^{t-1} (1-\gamma)^{t-1-j} \left(\epsilon_{i,j} + \Delta\beta_i(j)\right). \tag{70}$$

If

$$0 \leq \epsilon_{i,t} \leq \epsilon_{\max}, \qquad 0 \leq \Delta\beta_i(t) \leq \bar{\Delta}_{\beta,i}, \tag{71}$$

then the closed-loop practical safety lower bound is

$$\inf_{t \geq 0} H_{i,t} \geq -\frac{\epsilon_{\max} + \bar{\Delta}_{\beta,i}}{\gamma}. \tag{72}$$

This result shows that cross-sampling semantic-margin updates consume additional safety margin, and their influence is bounded by the maximum positive increment $\bar{\Delta}_{\beta,i}$. A smaller $\bar{\Delta}_{\beta,i}$ produces smoother semantic-boundary changes and a tighter closed-loop safety lower bound. When $\beta_{i,t} \equiv 0$ and $w_{i,t}^{\text{side}} \equiv 0$, the proposed semantic framework degenerates to the baseline hierarchical MPC-ECBF structure without semantic terms, and the bound becomes

$$\inf_{t \geq 0} h_{\text{EE}}(X_t) \geq -\frac{\epsilon_{\max}}{\gamma}. \tag{73}$$

V. EXPERIMENTAL VALIDATION

A. Verification Settings and Evaluation Metrics

The experimental validation is designed to answer three questions. First, it verifies whether the proposed semantic margin produces reasonable category- and context-dependent safety boundaries. Second, it evaluates whether the guarded semantic-margin update is necessary for maintaining MPC feasibility and smooth safety-boundary variation. Third, it examines whether the complete SEESM-based MPC-SECBF framework improves safety while preserving planning efficiency and real-time feasibility in dynamic social navigation scenarios.

All simulation experiments will be conducted in a C++/ROS or ROS/Gazebo environment using the same robot model, start and goal positions, obstacle trajectories, prediction horizon, MPC parameters, and computing platform for all compared methods. Unless otherwise stated, each method will be evaluated over 100 repeated trials in each simulation scenario. The real-world experiments will be conducted on a Scout 2.0 or equivalent autonomous mobile robot platform equipped with onboard LiDAR, IMU, and computing resources. Each real-world method configuration will be repeated at least 10 times under the same task setting.

The following metrics are used throughout the experiments. The success rate R_{suc} is the percentage of trials completed without collision or deadlock. The minimum obstacle distance $D_{\min}$ measures the smallest robot-obstacle separation during navigation. The semantic clearance D_{sem} is defined as the minimum value of $h_{\mathrm{SEE}}(X_t;\beta_{i,t})$ over all obstacles and sampling instants, and is used to evaluate whether the semantic safety boundary is satisfied. The path length L_{path} measures navigation efficiency. The control effort E_{cost} is computed by integrating the squared control input along the executed trajectory. The velocity variation $\Phi_{v|\omega}$ evaluates trajectory smoothness from linear and angular velocity changes. The average computation time T_{comp} measures planning and optimization efficiency. The MPC feasibility rate, average CBF slack, and semantic-margin increment $\Delta\beta$ are further reported to quantify the influence of semantic constraints on optimization feasibility and boundary smoothness.

B. Component-Level Validation

Component-level validation isolates the main modules of SEESM before testing the full navigation system. These experiments focus on semantic-margin generation, guarded margin updates, and closed-loop practical safety.

1) Semantic Margin Generation Test: This test verifies whether the semantic safety margin follows the intended category and context ordering. Six obstacle categories are considered: child, cyclist, vehicle, pedestrian, unknown object, and box. For each category, four interaction contexts are constructed: head-on approach, crossing motion, moving-away motion, and crowded local interaction. The generated semantic margin $\beta_{i,t}$ is recorded together with the category upper bound $\bar{\beta}_i$, head-on factor $f_{\mathrm{head},i}$, normalized time-to-collision $TTC_{\mathrm{norm},i}$, and normalized local density $\rho_{\mathrm{norm},i}$.

The expected evidence is that semantically sensitive objects, such as children and cyclists, receive larger margins than non-social objects under comparable geometric conditions, and that the margin increases under head-on, short-TTC, and crowded contexts. The result should be reported as a category-context matrix of $\beta_{i,t}$ and as time-series curves showing how the margin changes when the interaction context changes.

2) Guard Mechanism Ablation: This ablation evaluates whether the guarded update mechanism is necessary for feasibility and smooth safety-boundary evolution. The following variants are compared: full SEESM, category-only margin, context-only margin, SEESM without positive-increment limiting, SEESM without admissible-set projection, and SEESM without the MPC feasibility guard. All variants use the same perception input, obstacle trajectories, MPC horizon, and cost weights.

The reported quantities include the maximum positive margin increment $\max_t \Delta\beta_i(t)$, the MPC infeasibility rate, the average CBF slack, and the minimum closed-loop semantic safety quantity $H_{i,t}$. A useful ablation outcome should show whether removing rate limits causes abrupt semantic-boundary changes, whether removing admissible-set projection allows $\beta_{i,t}$ to exceed the available dynamic geometric margin, and whether removing the feasibility guard increases solver infeasibility.

3) Closed-Loop Practical Safety Check: This test validates the practical safety bound derived in the closed-loop analysis. During closed-loop execution, the quantity $H_{i,t} = h_{\mathrm{SEE}}(X_t;\beta_{i,t})$ is recorded for each obstacle and compared with the theoretical lower bound $-(\epsilon_{\max} + \bar{\Delta}_{\beta,i})/\gamma$. The positive semantic increment limit $\bar{\Delta}_{\beta,i}$ is varied over several values while keeping the same task and obstacle trajectories.

The expected presentation is a set of time-series plots for $H_{i,t}$, $\beta_{i,t}$, $\Delta\beta_i(t)$, and CBF slack. The experiment should report whether the measured minimum of $H_{i,t}$ remains above the derived bound and whether smaller $\bar{\Delta}_{\beta,i}$ produces smoother semantic-boundary variation and a tighter empirical safety margin.

C. Simulation Scenario Benchmarking

1) Experimental Setup: Simulation benchmarking evaluates the complete SEESM-based MPC-SECBF framework against representative baselines. The compared methods are: CESM-MPC-CBF, which uses only conventional Euclidean distance constraints; MPC-DCBF, which considers predicted dynamic obstacle states; EESM-MPC-ECBF, which uses the velocity-aware EESM without semantic margins; SEESM without the full guard mechanism; and the proposed full SEESM-MPC-SECBF.

Four simulation scenarios are used. Scenario 1 contains a single head-on pedestrian or cyclist and evaluates whether semantically sensitive obstacles trigger earlier and smoother avoidance. Scenario 2 contains crossing obstacles with mixed categories, including pedestrian, cyclist, and box, and tests whether different semantic objects induce different safety boundaries under comparable geometric risk. Scenario 3 is a crowded dynamic scene with three to six moving obstacles, including child, unknown object, and vehicle, and evaluates the effect of local crowding on safety and feasibility. Scenario 4 is a stress test with higher obstacle speed, higher obstacle density, or shorter time-to-collision, and is used to compare the feasibility and safety robustness of full SEESM against unguarded semantic constraints.

For each scenario, the main table should report $R_{\rm suc}$ $\uparrow$, $D_{\min}$ $\uparrow$ (m), $D_{\rm sem}$ $\uparrow$, $L_{\rm path}$ $\downarrow$ (m), $\Phi_{v|\omega}$ $\downarrow$, $T_{\rm comp}$ $\downarrow$ (ms), and MPC feasibility rate $\uparrow$ (%). Representative trajectory plots should compare the geometric paths of all methods. Time-series plots should show $\beta_{i,t}$, $H_{i,t}$, MPC slack, and computation time for the most challenging scenario.

2) Results and Discussion: The simulation results should be discussed according to the evidence needed for each claim. If full SEESM obtains a larger $D_{\rm sem}$ and $D_{\min}$ than EESM-MPC-ECBF under mixed-category scenarios, the result supports the value of semantic boundary modulation. If full SEESM maintains a higher MPC feasibility rate and lower slack than the unguarded SEESM variant, the result supports the necessity of the guarded semantic-margin update. If full SEESM maintains $T_{\rm comp}$ within the control-cycle deadline, the result supports real-time applicability.

Failure cases should also be reported. CESM-MPC-CBF is expected to fail or become overly aggressive when relative velocity and short-TTC interactions dominate the risk. Unguarded SEESM is expected to become overly conservative or infeasible when semantic margins increase abruptly. These cases should be illustrated with trajectory snapshots, margin time-series, and infeasibility records rather than only with aggregate metrics.

D. Real-World Verification

1) Experimental Setup: Real-world verification tests whether the proposed planner can run on a physical AMR and produce executable semantic-aware avoidance behavior. The platform is a Scout 2.0 or equivalent differential-drive AMR equipped with onboard LiDAR, IMU, and a computing unit. The robot performs approximately 20 m navigation tasks in outdoor or indoor dynamic environments.

Three real-world scenarios are recommended: a single pedestrian crossing the robot path, a mixed pedestrian-cyclist passing scenario, and a pedestrian-box mixed scenario. The semantic category can be provided by an online semantic detector if available. If semantic recognition is not part of the contribution, the categories can be provided by manual annotation or an experimental script; in that case, the paper should state explicitly that the experiment evaluates the SEESM planner rather than semantic classification accuracy.

At least two methods should be compared in the real-world tests: EESM-MPC-ECBF and the proposed full SEESM-MPC-SECBF. If safety conditions allow, an unguarded SEESM variant can be tested at low speed with a safety operator, but it should not be used in risky close-interaction trials. The reported metrics are $D_{\min}$, $D_{\rm sem}$, $L_{\rm path}$, $E_{\rm cost}$, $\Phi_{v|\omega}$, success rate, and average control frequency.

2) Results and Discussion: The real-world results should focus on whether SEESM changes the robot's behavior in a semantically meaningful and executable way. In the pedestrian and cyclist scenarios, the proposed method should be evaluated by whether it increases semantic clearance and avoids abrupt control changes. In the pedestrian-box scenario, the result should show whether the robot maintains a larger margin for the social obstacle while avoiding unnecessary detours around the non-social object.

The final presentation should include trajectory overlays, selected snapshots of the robot-obstacle interaction, and a quantitative table. The discussion should remain bounded: real-world verification demonstrates platform feasibility and semantic-aware behavior under the tested scenarios, while broader generalization to all social navigation environments requires additional evaluation with more diverse semantic perception inputs.

VI. Conclusion

Acknowledgment